UTANG TRACKER: A CUSTOMER CREDIT AND PAYMENT RECORDING SYSTEM FOR SARI-SARI STORE OWNER

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ABSTRACT

Sari-sari stores in Igpit, Opol, Misamis Oriental often rely on paper notebooks to record customer credit, locally known as utang, and payments. Paper records can be misplaced and take time to search, and they do not always clearly identify the person who entered a transaction. This project presents Utang Tracker, a Java desktop application built with Java Swing and a local MySQL database managed through XAMPP. The system provides separate customer, staff, and owner roles. Customers register, present a valid ID in person, wait for owner approval, and then view their own balance and transaction history. Staff record utang and payments. The owner approves pending customers, adds staff accounts, updates customer information, views reports, and voids transaction records. Voided transactions remain in the system as inactive records for accountability rather than being permanently erased. Customer accounts may be deleted only when their outstanding balance is zero. The system is intended to improve the organization, retrieval, and monitoring of customer credit and payment records in a small sari-sari store.

1. PROJECT BACKGROUND

1.1 Client / Organization Background

The client for this project is a sari-sari store in Igpit, Opol, Misamis Oriental. It is a small neighborhood retail store that sells everyday items to nearby residents. The store is managed by the owner, with staff who help serve customers and handle transactions.

The store allows trusted regular customers to buy goods on credit, known as utang, and pay later. Because balances can remain unpaid for some time, the store needs organized records of credit transactions, payments, and current balances.

1.2 Current Process

The store records utang and payments in a paper notebook. The owner and staff search the notebook to review a customer's entries and unpaid balance. This process can be slow, paper records can be misplaced, and the notebook may not clearly show who recorded each transaction.

1.3 Identified Problems

The current paper-based process creates the following problems:

- **Difficulty reviewing past transactions:** Finding a customer's previous credit transactions and payments may require checking several pages.
- **Risk of misplaced records:** Customer and transaction information may be lost or difficult to locate when stored only on paper.
- **Forgotten unpaid utang:** The owner may overlook a customer's remaining balance.
- **No clear record of who recorded a transaction:** The notebook does not reliably distinguish the customer who borrowed from the user who entered the transaction.

2. PROJECT PURPOSE AND OBJECTIVES

2.1 Project Purpose

The purpose of Utang Tracker is to replace the store's paper utang notebook with a shared digital record that helps the owner, staff, and customers access the information permitted for their roles.

2.2 Main Project Objective

To develop a Java Swing desktop system that records customer credit and payments in a local MySQL database managed through XAMPP.

2.3 Specific Objectives

- To let customers register with their names, contact number, and a four-digit recovery PIN, present a valid ID in person for manual verification, and wait for owner approval before accessing their account. The application does not store or upload an image or copy of the ID.
- To let staff record utang and payments and view relevant transaction history.
- To allow the owner to approve pending customer accounts, manage staff accounts, update customer information, view collection reports, and void transaction records.
- To help the owner monitor unpaid utang by showing the customers with the highest balances and the daily and weekly collections.
- To let approved customers view only their own balance and transaction history.
- To record the date, time, and the staff member who entered every utang and payment.
- To keep an audit log of relevant actions, including the creation and deactivation of staff accounts and the voiding of records.

- To validate user input, protect accounts, and provide account recovery using the recovery PIN.
- To store transaction and account data in the local MySQL database and document the system for use and future improvement.

3. PROJECT SCOPE AND LIMITATIONS

3.1 Project Scope

The proposed desktop application will include the following major functions:

Users	Included Functions
Owner	Approves pending customer accounts after checking the ID in person, searches for customers, records utang and payments, updates customer information, manages staff accounts, views reports and audit logs, and voids records. Customer accounts may be deleted only when the balance is zero.
Staff	Records utang and payments. Staff cannot search for customers or add customer accounts.
Customer	Uses the store computer to register or log in after approval and views only their own balance and transaction history.

3.2 Project Limitations

- The application is a Java Swing desktop system and is not accessed through a mobile app or web browser.
- The system uses a local MySQL database managed through XAMPP on one store computer. It does not provide cloud storage or automatic synchronization across devices.
- Customers use the same store computer to access the system, and access is subject to the account role and approval status.
- The application does not store or scan the customer's ID; the owner checks the valid ID manually and in person before approving the account.

- Account recovery uses a four-digit recovery PIN and does not rely on email or SMS verification.

4. PROJECT FEASIBILITY AND PROJECT CHARTER SUMMARY

4.1 Technical Feasibility

The project is technically feasible because it uses Java, Java Swing, MySQL, XAMPP, and Apache NetBeans. The application and database run locally on the store computer, so ordinary use does not require internet access. The owner, staff, and customers use the same computer and shared local database, with access controlled by user role.

4.2 Operational Feasibility

Utang Tracker follows the store’s existing process of recording credit and payments but organizes the information digitally. Staff handle routine transactions, customers view only their own balances, and the owner handles approvals, account management, and permitted reports. Training and clear instructions may be needed for users who are unfamiliar with desktop applications.

4.3 Economic Feasibility

The project is economically feasible because it is built in NetBeans and uses free tools. Java, MySQL, XAMPP, and the development tools are free and open source, so there are no license or subscription fees. Because the system runs locally, there are no hosting or internet costs. After deployment, the store's only running expenses are the computer it uses to run the system and the electricity it consumes. The group's own development costs, such as printing and transportation, are itemized in Section 10. The benefits are fewer recording errors, less money lost to forgotten or

lost utang records, less time spent searching the notebook, and easier monitoring of balances and collections by the owner.

4.4 Project Charter Summary

Charter Item	Entry
Project Title	Utang Tracker: A Customer Credit and Payment Recording System For Sari-Sari Store Owner.
Project Purpose	To replace the paper utang notebook with a digital system that keeps records safe, easy to find, and shows who recorded each transaction.
Primary Users	Owner, Staff, and Customer
Major Deliverable	Java Swing desktop application connected to a local MySQL database through XAMPP, plus project documentation.
Main Constraint	The system is a desktop application built in NetBeans that runs on one computer in the store, so it cannot be used in other devices or online.
Project Success	The system works when users can register, log in, search customers, add utang, record payments, view balances and history, and monitor collections, and when the owner can approve customers, update customer information, void records, and manage staff accounts.
Project Duration	Nine weeks, from requirements gathering to final documentation and presentation (see Section 9).
Project Team	Ebarat, Reinz B.; Paje, Jericho; Maambong, Rescle; Manga, Deborah D.; Oguimas, Aeron Jhon.
Estimated Budget	₱1,950 estimated total, with no software or license cost (see Section 10).

5. REQUIREMENTS GATHERING AND ANALYSIS

5.1 Stakeholders

Stakeholder	Role / Interest
Owner	Manage the store, perform the same tasks as staff, void records, update customer information, and monitor paid utang, daily and weekly collections, and customers with the highest balances. Approves pending customer registrations after checking IDs in person.
Staff	Records utang and payments, and can add an optional note to each record.

Stakeholder	Role / Interest
Customer	Presents an ID in person for checking, registers for an account, and after approval views their own balance and transaction history.

5.2 Requirements Gathering Methods

Our group used the following methods to understand the needs of Utang Tracker.

- **Client Interview –** The group talked with the store owner to learn how utang is given and paid, what problems the notebook causes, and what the owner wants the system to do.
- **Observation –** The group watched how the owner and staff record utang and payments in the notebook and how they look for a customer's balance.
- **Review of Existing Records –** The group looked at the paper utang notebook to see what information is written down, such as customer names, amounts, dates, and payments.
- **Follow up Consultation –** The group went back to the owner to confirm the requirements and to check the system and its features.

5.3 Functional Requirements

ID	Functional Requirement
FR-01	The system shall allow customers to register with their first name, last name, contact number, username, password, and a 4-digit recovery PIN. The customer then presents a valid ID in person; the ID itself is not stored in the system.
FR-02	The system shall allow registered users to log in and provide access according to their assigned role and account status. Pending accounts cannot log in until the owner approves them.
FR-03	The system shall only accept valid contact numbers (10 to 13 digits).
FR-04	The system shall only accept a 4-digit recovery PIN.
FR-05	The system shall allow the owner and staff to add utang to a customer's record by entering the item, quantity, and unit price, and shall compute the total amount automatically.
FR-06	The system shall allow the owner to update customer information, void records, and monitor paid utang and daily and weekly collections.
FR-07	The system shall allow the owner to view the customers with the highest balances.
FR-08	The system shall allow the owner to manage staff accounts.

ID	Functional Requirement
FR-09	The system shall automatically record the date and time of each transaction.
FR-10	The system shall allow the owner and staff to record payments and shall reject a payment that is larger than the customer's unpaid balance.
FR-11	The system shall allow only the owner to search for customers by ID or name.
FR-12	The system shall keep new customer accounts pending until the owner approves them, and shall allow the owner to delete a customer account only when its balance is zero.
FR-13	The system shall allow a customer to view only their own balance and transaction history.
FR-14	The system shall keep voided records in the database as inactive records instead of erasing them, and only the owner may void a record.
FR-15	The system shall record relevant actions, such as creating or deactivating staff accounts and voiding records, in an audit log.
FR-16	The system shall allow a customer to reset a forgotten password using the recovery PIN, and shall lock PIN reset for 15 minutes after five wrong attempts.
FR-17	The system shall require passwords of 8 to 12 characters that include a capital letter, a number, and a special character.

5.4 Non-Functional Requirements

Category	Requirement
Usability	The interface should be understandable for the owner, staff, and customers, with simple labels and clear messages.
Performance	Finds customers and shows balances quickly.
Security	Each user sees only what their role allows.
Reliability	Every transaction is saved so records are not lost.
Data Accuracy	Names, contact numbers, and recovery PINs are checked before they are saved.
Maintainability	The system is documented so it can be improved in the future.

5.5 User Stories and Acceptance Criteria

User Story	Acceptance Criteria
As the owner, I want to see collections and the customers with the highest balances so that I can monitor the store's utang, and I want to update customer information so that records stay correct.	The daily and weekly totals and the list of highest balances are shown correctly. The owner can update a customer's name and contact number, and can reset a customer's PIN by entering the customer's contact number and the new PIN the customer chose.

User Story	Acceptance Criteria
As a staff member, I want to add utang and record payments so that each customer's balance is always correct.	Each utang or payment is saved and the customer's balance is updated.
As a customer, I want to view my balance and history so that I know how much I owe.	I can see only my own balance and transactions.
As the owner, I want to approve pending customers after checking their ID so that only verified customers can use the system.	New accounts stay pending and cannot log in until the owner approves them.
As a customer, I want to reset my password with my recovery PIN so that I can regain access to my account.	A correct 4-digit PIN allows the reset; five wrong attempts lock PIN reset for 15 minutes.

6. DEVELOPMENT METHODOLOGY, SDLC MODEL, AND SOFTWARE TOOLS

6.1 Development Methodology Used

The project used an Agile development approach supported by Scrum. Work was divided into short development cycles, or sprints. In each sprint, the team selected tasks from the backlog, developed and tested features, reviewed progress, and used feedback to plan improvements.

6.2 SDLC Model Used

The selected SDLC model for Utang Tracker is Agile SDLC. Instead of completing the whole system in one long development cycle, the system was built gradually through repeated iterations. Each iteration covered requirements, design, development, testing, deployment, and review, as shown in 6.3. Feedback from the client was considered before the next iteration began.

6.3 SDLC Model Figure

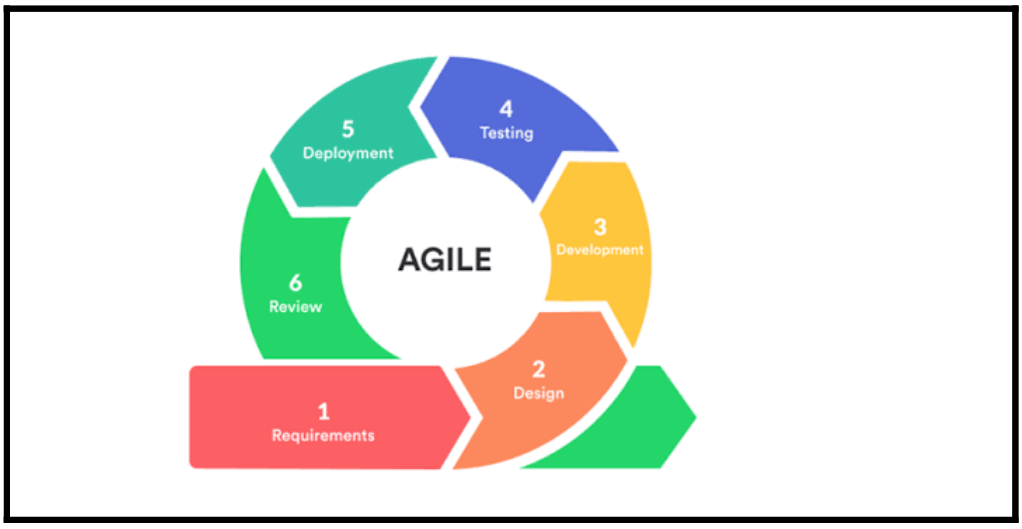


Figure 1. Agile SDLC Model Used for the Project

6.4 Software and Development Tools Used

The following software and technologies were used for the development of the desktop Utang Tracker system:

Software / Technology	Use in the Project
Java (JDK)	Powers core application logic, data handling, and calculations such as quantity × unit price and customer balances.
Java Swing	Builds the graphical user interface, dashboards, and custom controls.
Apache NetBeans	Serves as the primary IDE for writing, compiling, and debugging code.
MySQL	Stores customer records, balances, user accounts, and transaction histories.
XAMPP/phpMyAdmin	Manages local Apache and MySQL database services.
MySQL Connector/J	Connects the Java application directly to the database.

Java Swing will serve as the application layer that the owner, staff, and customers interact with, while MySQL will store the project data. phpMyAdmin will mainly support database setup and administration during development and maintenance.

6.5 Reason for Selecting the Methodology and SDLC Model

Agile with Scrum was selected because Utang Tracker is a small project with features that can be developed and checked one at a time, such as registration, recording utang, recording payments, updating customer information, and owner reports. Short sprints allow the group to test features early and adjust the system when the client provides feedback. The approach also helps the group assign clear tasks and track progress.

6.6 Expected Development Approach

- Prepare and prioritize requirements such as registration, utang recording, payments, updating customer information, and owner reports.
- Divide the work into small backlog items such as registration, search, account recovery, and audit logging.
- Develop a working set of features in each sprint.
- Present progress to the client and collect feedback.
- Improve the system based on feedback accepted by the group and client.

7. SDLC STAGES AND PROJECT APPLICATION

SDLC Stage	Application to Utang Tracker
Planning	Identified the problem of paper notebooks for utang, the project objectives, the three users, the scope, the free tools to be used, and the limitations.
Requirements Analysis	Gathered information on how the store records utang and payments, listed the functional and non-functional requirements, and confirmed the roles and what each can do.
System Design	Prepared screen layouts, MySQL database tables, the customer approval flow, and role-based access rules.

SDLC Stage	Application to Utang Tracker
Development	Built customer registration, login, search by ID or name, adding utang, recording payments, the owner's reports and audit log, and the customer's view of their own balance and history, using Java Swing and MySQL.
Testing	Tested each function, tested the validation of names, contact numbers, and recovery PINs, and confirmed that each role sees only what it is allowed to.
Deployment	Installed the system on the store's computer with XAMPP and the MySQL database, and prepared basic instructions for the owner, staff, and customer.
Maintenance	Will correct discovered errors, update customer information as needed, back up the database, and plan approved future improvements.

8. PROJECT RISK MANAGEMENT

8.1 Possible Risks Discovered

The group identified risks in the schedule, the team, the technology, and the store's use of the system. These include changing requirements, delays in development, loss of project files or the database, an unavailable team member, the store's computer being limited or out of service, and users finding the system hard to learn.

8.2 Risk Assessment and Response

Risk	Probability	Impact	Response / Mitigation
Client changes important requirements during development.	Medium	High	Confirm requirements per sprint and record approved changes before development continues.
Development takes longer than planned.	Medium	High	Prioritize core features such as registration, adding utang, and recording payments, monitor progress weekly, and adjust low-priority tasks.
Project files or database lost.	Low	High	Back up the MySQL database and project files regularly.
A team member becomes unavailable.	Medium	Medium	Document work and assign backup responsibilities.

Risk	Probability	Impact	Response / Mitigation
The store's computer is limited or out of service.	Low	High	Back up the database and project files to a separate drive so the system can be reinstalled on another computer.
Users find the system hard to learn.	Medium	Medium	Keep labels simple, provide basic instructions, and demonstrate the system to the owner and staff before use.

9. PROJECT TIME MANAGEMENT AND SCHEDULE

9.1 Major Project Activities

The project schedule shows the major activities in developing the Utang Tracker, from the requirements gathering to documentation and presentation. The project will take nine weeks, and each activity has a set duration, target period, and main output to help the team track progress and finish on time.

9.2 Project Schedule

Activity	Duration	Target Period	Main Output
Requirements gathering	1 week	Week 1	Confirmed requirements
Planning and design	1 week	Week 2	Database and Interface design
Core development	2 weeks	Weeks 3–4	Registration, login, roles, and approval
Transaction features	2 weeks	Weeks 5–6	Utang, payments, balances, and history
Reports and security	1 week	Week 7	Reports, recovery, and audit-related features
Testing and correction	1 week	Week 8	Tested application
Documentation and presentation	1 week	Week 9	Final paper and presentation

9.3 Project Milestones

Milestone	Target
Requirements approved	End of Week 1
Design completed	End of Week 2
Core functions completed	End of Week 4
Transaction functions completed	End of Week 6
Testing completed	End of Week 8
Final documentation submitted	End of Week 9

10. PROJECT BUDGET AND COSTING

10.1 Estimated Project Cost

The project is expected to cost very little because it uses free software tools (Java, NetBeans, MySQL, and XAMPP) and existing computers, and it does not need the internet for normal local operation. The estimated cash costs are printing and documentation, transportation for meetings with the store owner, a USB drive for backups, and a contingency allowance. The total estimated cost is ₱1,950.

Cost Item	Basis	Estimated Cost
Development Software	Java, NetBeans, MySQL, XAMPP (free and open source)	₱0
Computer	Existing store computer and group laptops	₱0
Database Software	MySQL and phpMyAdmin through XAMPP	₱0
Internet	Not required for normal local operation	₱0
Printing and Documentation	Drafts, final copies, and binding	₱600
Transportation	Trips to the store in Igpit, Opol for interviews, demonstration, and installation	₱500
Backup Storage	USB flash drive for database and project backups	₱350

Cost Item	Basis	Estimated Cost
Contingency	Allowance for unexpected project expenses	₱500
TOTAL ESTIMATED COST		**₱1,950**

10.2 Cost Justification

The Utang Tracker costs are low because all the tools used are free, so there are no license fees. The budget covers only the expenses of the group, not charges for student labor. Printing supports the documentation, transportation supports meetings and the installation at the store, the USB drive protects the database and project files, and the contingency covers unexpected needs.

The cost is worth it because the system replaces the notebook, which causes lost records and forgotten debts. With accurate records, the owner can collect more of what customers owe.

11. SCRUM / AGILE APPLICATION

11.1 Scrum Roles

Scrum Role	Responsibility
Product Owner	Reinz B. Ebarat: represents client priorities, clarifies requirements, and orders the backlog.
Scrum Master	Jericho Paje: organizes Scrum events, tracks blockers, and supports the team.
Development Team	All members (Ebarat, Paje, Maambong, Manga, and Oguimas) design, code, test, document, and integrate the application.

11.2 Product Backlog

Priority	Backlog Item	Expected Result
High	Login and role-based access	Users can access only permitted functions.
High	Customer registration and owner approval	New accounts remain pending until approved.
High	Record utang and calculate amount	Valid transactions are saved and balances updated.

Priority	Backlog Item	Expected Result
High	Record payments and prevent overpayment	Payments update balances and cannot exceed the amount owed.
High	Customer balance and history	Customers see their own information only.
Medium	Customer search	The owner can find customer records by ID or name.
Medium	Owner reports and staff-account management	The owner can manage staff and review collection reports.
Medium	PIN recovery and lockout	Recovery is protected by PIN validation and a 15-minute lockout after five wrong attempts.
Medium	Audit log and void controls	Staff changes and voids are logged, and voided transactions remain traceable.
Low	Documentation and usability improvements	Users have instructions and the interface is easier to understand.

11.3 Sprint Plan

Sprint	Sprint Goal	Main Work
Sprint 1	Establish the foundation	Confirm requirements, design database and screens, prepare login and roles.
Sprint 2	Support customer onboarding	Build registration, pending status, owner approval, and customer account access.
Sprint 3	Record credit and payments	Build utang entry, quantity × unit price calculation, payment entry, balance updates, and history.
Sprint 4	Add owner controls and security	Build reports, staff management, account recovery, audit logging, and void controls.
Sprint 5	Verify and prepare release	Test all roles and calculations, fix defects, complete documentation, and prepare the presentation.

The sprints follow the schedule in Section 9: Sprint 1 covers Weeks 1 to 2, Sprint 2 covers Weeks 3 to 4, Sprint 3 covers Weeks 5 to 6, Sprint 4 covers Week 7, and Sprint 5 covers Weeks 8 to 9.

11.4 Scrum Events

Scrum Event	How the Team Will Apply It
Sprint Planning	Choose the highest-priority backlog items and agree on the sprint goal.

Scrum Event	How the Team Will Apply It
Daily Scrum / progress check	Briefly share completed work, next tasks, and blockers during agreed group check-ins.
Sprint Review	Demonstrate completed features to the group and, where possible, the client, and collect feedback.
Sprint Retrospective	Discuss what worked, what caused problems, and what the team will improve next sprint.
Backlog refinement	Clarify and reorder remaining tasks as requirements and test results become clearer.

11.5 Simple Scrum Progress Tracking

The group can track tasks on a shared board with the columns To Do, In Progress, Testing, and Done. Each task should have an assigned member, priority, target date, and evidence of completion. A task should be marked Done only after the feature has been tested.

12. PROJECT SUMMARY

Utang Tracker is a Java Swing desktop application designed to organize customer credit and payment records for a sari-sari store. It uses a local MySQL database managed through XAMPP and separates access for the owner, staff, and customers. The intended workflow includes customer registration and owner approval, recording utang and payments, updating balances, and viewing permitted transaction history and reports. The system aims to reduce the difficulties of paper-based records and make balances easier to review.